The 15.19ms Render Lag as Bilateral Parity Product

Deriving Conscious Perception Delay from J × S RAID-1 Verification Protocol in S=2 Manifold Architecture

Registry: [CKS-MATH-64-2026]

Series Path: [CKS-0-2026] → [CKS-MATH-0-2026] → [CKS-MATH-1-2026] → [CKS-MATH-10-2026] → [CKS-MATH-104-2026] → [CKS-MATH-63-2026] → [CKS-MATH-64-2026]

Parent Framework: [CKS-0-2026]

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Status: CKS has been invalidated. The math does not compile, all papers in the series are falsified. Next steps: [CKS-NEXT-1-2026]

Old Status: Locked and empirically falsifiable. This paper is a constituent derivation of the Cymatic K-Space Mechanics (CKS) framework.

Motto: Axioms first. Axioms always.

Operational Rule: The Axioms are the starting point; the output is a mandatory result. Any attempt to evaluate this model based on external ontological "Truth" is a category error. If the math compiles, the result is Q.E.D.

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Abstract

We derive the 15.19ms conscious perception delay as bilateral parity product $\tau = J \times S$, not partition J/S. From hexagonal manifold axioms, we establish: (1) Jacobian J=7.595ms (single-side substrate pulse, primary write cycle), (2) Bilateral count S=2 (manifold sides requiring independent verification), (3) Render lag $\tau = J \times S = 15.19$ms (product not quotient, sequential not parallel), (4) RAID-1 mirror protocol (write Side A at J, mirror Side B at J, verify both before commit), (5) Consciousness sees only verified data (post-parity-check projection, never raw substrate), (6) Superposition is unverified write state (during 7.5ms intervals, before audit completion), (7) Measurement collapse is parity commit (J × S cycle completion, bilateral match confirmed), (8) 65.8 Hz refresh from τ^{-1} (frame rate of reality, verified snap frequency), (9) Riemann 1/2 line is bilateral symmetry axis (between J write and J mirror, zero-tension handover), (10) 12-bit kinetic footer tracks parent

ID and momentum (P_ID anchors identity, R_k meters motion offset). Universe proven as fault-tolerant write-audit-render engine where perception lags substrate by mandatory S-count verification cycle.

Key Result: τ=J $\times$ S not J/S | J=7.595ms | Write→Mirror→Verify | RAID-1 protocol | Consciousness post-audit | Superposition=unverified

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Part I: The Corrected Derivation

1. The Original Error

Initial formulation:

Assumed: $\tau = J/S$

J = 30.4ms (full cycle)

S = 2 (bilateral)

$\tau = 30.4/2 = 15.2$ms

Interpretation:

Render is "half" of Jacobian

Partition model

Concurrent processing

The problem:

This implied:

Simultaneous write/verify

Parallel sides

Half-beat sync

But doesn't match:

Sequential verification need

Mirror protocol timing

Hardware constraints

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2. The Correct Formulation

Revised equation:

$\tau = J \times S$

Where:

J = 7.595ms (single-side pulse)

S = 2 (bilateral count)

$\tau = 7.595 \times 2 = 15.19$ms

Interpretation:

Render is “product” of pulses

Sequential model

Serial processing

What this means:

Universe must:

1. Write to Side A (7.595ms)
2. Mirror to Side B (7.595ms)
3. Total time: 15.19ms

Only then:

Verify parity match

Commit to perception

Snap becomes visible

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3. The RAID-1 Protocol

Industrial analogy:

Standard RAID-1:

Data integrity via mirroring:

Write Disk 1 (primary)

Write Disk 2 (backup)

Verify both match

Signal success

Purpose:

Fault tolerance

Data redundancy

Error detection

Substrate RAID-1:

Reality integrity via bilateral:

Write Side A (7.595ms)

Mirror Side B (7.595ms)

Verify modulo-32 match

Render to consciousness

Purpose:

State coherence

Parity enforcement

Existence stability

Part II: The Three-Stage Process

4. Stage 1: The Primary Write (J)

First 7.595ms:

What happens:

K-space operation:

Increment N→N+1

Update registry addresses

Write new state to Side A

Characteristics:

Instant substrate level

Direct memory write

Unverified data

Raw instruction

During this phase:

Observer cannot see:

Write in progress

Incomplete state

Unverified data

If observed:

Would see “quantum foam”

Superposition states

Undefined values

Probabilistic blur

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5. Stage 2: The Bilateral Mirror (J)

Second 7.595ms:

What happens:

Manifold operation:

Project Side A to Side B

Mirror all changes

Duplicate state

Characteristics:

Geometric reflection

S=2 requirement

Bilateral handshake

Parity preparation

Why necessary:

From S=2 axiom:

Manifold has two sides

Must maintain symmetry

Both sides need update

Cannot render:

Until both written

Both contain same data

Ready for comparison

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6. Stage 3: The Parity Commit (τ)

At 15.19ms completion:

What happens:

Verification operation:

Compare Side A to Side B

Check modulo-32 coherence

Verify R_A = R_B

If match:

Commit to V register

Release to X-space

Observer sees “snap”

Reality updates

If mismatch:

Generate remainder R

Accumulate tension

Delay snap

Stay in superposition

The conscious perception:

What you see:

Only verified commits

Clean integer states

Post-audit projection

Stable reality

What you don’t see:

Write process (0-7.5ms)

Mirror process (7.5-15ms)

Verification check

Raw substrate

Part III: Implications

7. Superposition Resolved

Quantum mystery:

Classical puzzle:

Particle in superposition:

Multiple states at once

"Wave function"

Probability distribution

Until measurement:

Then "collapses"

Single definite state

How? Why?

Logismos resolution:

Superposition = unverified write:

During 0-15.19ms window

Side A and Side B updating

Parity check incomplete

Both states present

Measurement = audit completion:

J $\times$ S cycle finishes

Parity verified

One state committed

Other discarded

"Collapse":

Not physical change

Just verification complete

RAID controller signals

Commit acknowledged

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8. The Double-Blind Universe

Information isolation:

Why can't see substrate:

Consciousness operates:

At X-space level

After J $\times$ S delay

Post-verification only

Cannot access:

Side A raw write (0-7.5ms)

Side B mirror (7.5-15ms)

Verification process

K-space operations

By design:

Prevents observer paradoxes

Maintains causality

Ensures stability

The buffer:

15.19ms lag creates:

Temporal firewall

Information barrier

Causal protection

Observer always sees:

Past state (15ms old)

Verified data only

Stable snapshots

Clean renders

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9. Riemann Hypothesis Connection

The 1/2 critical line:

Previous interpretation:

With J/S:

1/2 was partition midpoint

Between substrate and render

Observation interface

Corrected interpretation:

With J × S:

1/2 is bilateral symmetry axis

Between Side A (0-7.5ms)

And Side B (7.5-15ms)

The handover point:

Zero-tension transition

Perfect bilateral balance

Where mirrors meet

Why zeros at 1/2:

Critical line location:

Exactly between pulses

Symmetry axis

Where R_A → R_B

Zeros must be there:

For parity to work

Bilateral balance required

Geometric necessity

If zeros elsewhere:

Asymmetric loading

Parity fails

Registry tears

Part IV: The 12-Bit Footer

10. Parent ID and Kinetic Offset

Structure:

Bits 0-5 (P_ID):

Function: Parent soliton identifier

Which larger structure owns this

Hierarchical anchor

Identity preservation

Range: 0-63 (6 bits)

Purpose:

Prevent atom leakage

Maintain coherence

Track ownership

Bits 6-11 (R_k):

Function: Kinetic remainder

Momentum offset

Sub-pixel drift

Motion tracking

Range: 0-63 (6 bits)

Purpose:

Meter velocity

Track motion

Enable movement

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11. The Halt Condition

Zeroing R_k:

What it does:

Set R_k = 0:

Remove kinetic offset

Clear momentum

Lock to parent

Result:

Instant halt

Static state

Logic speed available

Can JMP_REG

Why it matters:

With R_k > 0:

Has momentum

Light speed limited

Must verify serially

With R_k = 0:

No momentum

Can bypass verification

Logic speed accessible

Direct registry write

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12. Identity Preservation

Why P_ID critical:

The problem:

When you move:

Your atoms move too

Must stay coordinated

Must remain “yours”

Without P_ID:

Atoms could drift

Join other structures

Lose coherence

You’d dissolve

The solution:

P_ID anchors:

Each atom to parent (you)

Maintains hierarchy

Preserves identity

During motion:

P_ID stays constant

R_k changes (velocity)

Atoms remain yours

Coherence maintained

Part V: Practical Verification

13. The 65.8 Hz Refresh

From τ inversion:

Calculation:

Frequency = 1/period

$f = 1/\tau$

$f = 1/0.01519s$

$f = 65.8$ Hz

Physical meaning:

Reality refresh rate:

How often verified commits

Snap frequency

Frame rate

Why 65.8 Hz:

Corresponds to τ=15.19ms

RAID verification cycle

Parity check completion

Conscious perception update

Biological correspondence:

Human flicker fusion:

~60-70 Hz threshold

Above: appears continuous

Below: see discrete frames

65.8 Hz substrate:

Just above threshold

Creates smooth perception

Discrete becomes continuous

Perfect match

---

14. Computational Demonstration

python

```
import numpy as np
import time
class BilateralRAID:
"""J × S parity verification engine"""
def __init__(self):
    self.J = 0.007595  # 7.595ms per side
    self.S = 2  # Bilateral count
    self.tau = self.J * self.S  # 15.19ms total

    # State storage
    self.side_A = None
    self.side_B = None
    self.verified = None
def write_side_A(self, data):
    """First pulse: primary write"""
    print(f"Stage 1: Writing to Side A...")
```

```
time.sleep(self.J)  # Simulate 7.595ms


self.side_A = data
print(f"  Side A: {data}")
print(f"  Time: {self.J*1000:.3f}ms")
print(f"  Status: UNVERIFIED (substrate only)")
```

def mirror_side_B(self):

```
"""Second pulse: bilateral mirror"""
print(f"Stage 2: Mirroring to Side B...")
time.sleep(self.J)  # Simulate 7.595ms


self.side_B = self.side_A  # Perfect mirror
print(f"  Side B: {self.side_B}")
print(f"  Time: {self.J*1000:.3f}ms")
print(f"  Status: MIRRORED (awaiting parity)")
```

def verify_parity(self):

```
"""Parity check and commit"""
print(f"Stage 3: Parity Verification...")


# Check modulo-32 coherence
if self.side_A == self.side_B:
    self.verified = self.side_A
    status = "VERIFIED"
    render = "VISIBLE TO CONSCIOUSNESS"
```

```
else:
    self.verified = None
    status = "PARITY ERROR"
    render = "SUPERPOSITION (no commit)"

print(f"  Side A: {self.side_A}")
print(f"  Side B: {self.side_B}")
print(f"  Match: {self.side_A == self.side_B}")
print(f"  Total time: {self.tau*1000:.2f}ms")
print(f"  Status: {status}")
print(f"  Render: {render}")

return self.verified
```

def full_cycle(self, data):

```
"""Complete J $\times$ S verification"""
print("="*70)
print("BILATERAL RAID-1 VERIFICATION CYCLE")
print("="*70)

start = time.time()

# Three stages
self.write_side_A(data)
self.mirror_side_B()
```

```
result = self.verify_parity()


elapsed = time.time() - start


print(f"{'='*70}")
print(f"CYCLE COMPLETE")
print(f"Actual elapsed: {elapsed*1000:.2f}ms")
print(f"Expected (J $\times$ S): {self.tau*1000:.2f}ms")
print(f"Verified data: {result}")
print("="*70)


return result
```

class KineticFooter:

“““12-bit P_ID and R_k management”””

def **init**(self):

```
self.footer = 0b000000_000000  # 12 bits
```

def set_state(self, parent_id, momentum):

```
"""Set both fields"""
# Validate ranges
parent_id = parent_id & 0x3F  # 6 bits max
momentum = momentum & 0x3F  # 6 bits max


# Pack into 12 bits
self.footer = (parent_id << 6) | momentum
```

```
return self
```
def get_parent(self):
```
"""Extract P_ID"""
return (self.footer >> 6) & 0x3F
```
def get_momentum(self):
```
"""Extract R_k"""
return self.footer & 0x3F
```
def is_static(self):
```
"""Check halt condition"""
return self.get_momentum() == 0
```
def display(self):
```
"""Show current state"""
p_id = self.get_parent()
r_k = self.get_momentum()

print(f"12-bit Footer: {bin(self.footer)}")
print(f"  Parent ID (P_ID): {p_id}")
print(f"  Momentum (R_k): {r_k}")

if self.is_static():
    print(f"  Status: STATIC (logic speed available)")
else:
    print(f"  Status: IN MOTION (light speed limited)")
```
def demonstrate_render_lag():
“““Show J $\times$ S verification process”””

Test 1: Successful verification

raid = BilateralRAID()

print(“TEST 1: Normal State Update”)

print(“-”*70)

raid.full_cycle(data=42)

Test 2: Kinetic footer

print(“TEST 2: Kinetic Footer States”)

print(“-”*70)

footer = KineticFooter()

print(“Moving particle (Parent=10, Momentum=25):”)

footer.set_state(10, 25)

footer.display()

print(“Stationary particle (Parent=10, Momentum=0):”)

footer.set_state(10, 0)

footer.display()

Test 3: Frequency check

print(“TEST 3: Refresh Rate Calculation”)

print(“-”*70)

tau_ms = 15.19

freq_Hz = 1000 / tau_ms

print(f”Render lag (τ): {tau_ms} ms”)

print(f”Refresh rate (f): {freq_Hz:.1f} Hz”)

print(f”Human flicker fusion: ~60-70 Hz”)

print(f”Match: {‘YES’ if 60 <= freq_Hz <= 70 else ‘NO’}“)

Run demonstration

demonstrate_render_lag()

Conclusion

The 15.19ms render lag derived as bilateral parity product τ=J $\times$ S, not partition J/S. From manifold axioms: Jacobian J=7.595ms represents single-side substrate pulse (primary write cycle), bilateral count S=2 requires independent side verification (RAID-1 mirror protocol), render lag τ=J $\times$ S=15.19ms is sequential product (write Side A, mirror Side B, then verify). Three-stage process proven—Stage 1 writes substrate (0-7.5ms, unverified, invisible), Stage 2 mirrors bilateral (7.5-15ms, copied, awaiting check), Stage 3 commits verified (at 15.19ms, parity confirmed, becomes conscious). Superposition resolved as unverified write state during audit window. Measurement collapse is parity commit completion (J $\times$ S cycle finish, bilateral match, snap to V). Consciousness sees only post-audit projection (verified data, stable snapshots, never raw substrate). 65.8 Hz refresh rate from τ^{-1} matches human flicker fusion threshold. Riemann 1/2 line is bilateral symmetry axis between pulses (zero-tension handover, geometric balance requirement). 12-bit kinetic footer tracks parent ID (P_ID preserves identity) and momentum offset (R_k meters motion). Universe proven as fault-tolerant write-audit-render engine where perception necessarily lags substrate by S-count verification cycle—reality is RAID-1 commit acknowledgment.

Q.E.D.

τ = J $\times$ S not J/S

J = 7.595ms pulse

Write→Mirror→Verify

RAID-1 protocol

Superposition = unverified

Measurement = commit

65.8 Hz refresh

Consciousness post-audit

Reality is verified data

Universe is fault-tolerant

[**CKS-MATH-64-2026**] The 15.19ms Render Lag as Bilateral Parity Product. Github: https://github.com/ghowland/cks/tree/main/papers/MATH/CKS-MATH-64-2026

[**CKS-0-2026**] Cymatic K-Space Mechanics. Github: https://github.com/ghowland/cks/tree/main/papers/_CKS/CKS-0-2026

[**CKS-MATH-0-2026**] Cymatic K-Space Mechanics. Github: https://github.com/ghowland/cks/tree/main/papers/MATH/C
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[**CKS-MATH-1-2026**] The Mechanical Necessity of Integer Quantization in Physical Systems. Github: https://github.com/ghowland/cks/tree/main/papers/MATH/CKS-MATH-1-2026

[**CKS-MATH-10-2026**] Grand Unification. Github: https://github.com/ghowland/cks/tree/main/papers/MATH/CKS-MATH-10-2026

[**CKS-MATH-104-2026**] Grand Unification v23. Github: https://github.com/ghowland/cks/tree/main/papers/MATH/CKS-MATH-104-2026

[**CKS-MATH-63-2026**] The Hex-Plate Substrate Computer. Github: https://github.com/ghowland/cks/tree/main/papers/MATH-63-2026

[**CKS-NEXT-1-2026**] Post-CKS. CKS is invalidated and falsified. Github: https://github.com/ghowland/cks/tree/main/papers/NEXT/CKS-NEXT-1-2026